GRADE 8

TERM 1

MATHEMATICS

STUDY NOTES

THINK

DIGITAL ACADEMY

1. Whole Numbers

#1

Ordering and Comparing Whole Numbers

Whole Numbers

- **Whole Numbers** are a set of numbers including all positive integers and 0.
- **Whole Numbers** are a part of real numbers that do not include fractions, decimal numbers, or negative numbers.
    - **Whole Numbers**
      $\mathbb{N}_0$= {0, 1, 2, 3, 4, ...}

Example: 673 584 231

Millions			Thousands			Units		
H	**T**	**U**	**H**	**T**	**U**	**H**	**T**	**U**
6	7	3	5	8	4	2	3	1

- If you want to **order and compare** whole numbers, you must compare the corresponding place values.
  Example:
  Compare the following whole numbers and arrange them from smallest to largest:
  22 560, 68 492, 65 500
  Step 1: Start by arranging the ten thousands → 20 thousand, 60 thousand and 60 thousand → 22 560 is the smallest
  Step 2: Arrange the thousands (comparing 65 500 and 68 492) → 8 thousand and 5 thousand → 5 thousand is smaller, thus 65 500 is smaller.
  Step 3: Arrange from smallest to largest → 22 560; 65 500; 68 492

Symbols

- We use the following **symbols** when comparing whole numbers.

Symbol	Meaning	Example
$=$	Equal to	$15 + 13 = 28$
$>$	Greater than	$23 > 15$
$<$	Smaller than	$7 < 30$

Prime Numbers and Composite Numbers

- **Prime Numbers:** A prime number has only two factors, 1 and itself.
  e.g. 13 is a prime since it's only two factors are 1 and 13

- **Composite Numbers:** natural numbers with more than two different factors. Thus, these numbers can be divided exactly by at least one number that is not 1, by 1 and by the number itself.
  e.g. 4 is a composite number since it can be divided by 1, 2 and 4.

#2

Properties and Calculations of Whole Numbers

Addition

- Changing the order of the terms in the sum doesn't affect the answer.

$$\boxed{\boldsymbol{a + b = b + a}}$$

e.g. $12 + 8 = 20$
$8 + 12 = 20$
$\therefore 12 + 8 = 8 + 12 = 20$

- Grouped terms can be changed without affecting the result of your sum.

$$\boxed{\boldsymbol{(a + b) + c = a + (b + c)}}$$

e.g. $(12 + 8) + 5 = 20 + 5 = 25$
$12 + (8 + 5) = 12 + 13 = 25$
$\therefore (12 + 8) + 5 = 12 + (8 + 5) = 25$

- Adding zero to any number doesn't change the number (it is equal to that same number).

$$\boxed{\boldsymbol{a + 0 = 0 + a = a}}$$

e.g. $5 + 0 = 5$
$0 + 5 = 5$
$\therefore 5 + 0 = 0 + 5 = 5$

Multiplication

- Changing the order of the terms that you multiply doesn't affect the answer.

$$\boxed{\boldsymbol{a \times b = b \times a}}$$

e.g. $10 \times 7 = 70$
$7 \times 10 = 70$
$\therefore 10 \times 7 = 7 \times 10 = 70$

- Changing how numbers in a product are grouped doesn't affect the result.

$$\boxed{\boldsymbol{(a \times b) \times c = a \times (b \times c)}}$$

e.g. $(10 \times 7) \times 2 = 70 \times 2 = 140$
$10 \times (7 \times 2) = 10 \times 14 = 140$
$\therefore (10 \times 7) \times 2 = 10 \times (7 \times 2) = 140$

- Grouped added terms can have a multiplication spread across it.

$$\boxed{\boldsymbol{a(b + c) = ab + ac}}$$

e.g. $10(4 + 6) = 10(10) = 100$
$10(4) + 10(6) = 40 + 60 = 100$
$\therefore 10(4 + 6) = 10(4) + 10(6) = 100$

- Grouped subtracted terms can have a multiplication spread across it.

$$\boxed{\boldsymbol{a(b - c) = ab - ac}}$$

e.g. $5(20 - 10) = 5(10) = 50$
$5(20) - 5(10) = 100 - 50 = 50$
$\therefore 5(20 - 10) = 5(20) - 5(10) = 50$

- One multiplied by any number leaves the number unchanged (will be the number).

$$\boxed{\boldsymbol{1 \times a = a \times 1 = a}}$$

e.g. $1 \times 10 = 10$
$10 \times 1 = 10$
$\therefore 1 \times 10 = 10 \times 1 = 10$

Subtraction

- Changing the order of the terms that you are subtracting affects the final answer.

The order of the numbers cannot be rearranged

e.g. $20 - 10 = 10$
$10 - 20 = -10$
$\therefore 20 - 10 \neq 10 - 20$

- Changing how numbers being subtracted are grouped affects its result so the order can't be changed.

The order of the numbers cannot be rearranged

e.g. $(20 - 10) - 5 = 10 - 5 = 5$
$20 - (10 - 5) = 20 - 5 = 15$
$\therefore (20 - 10) - 5 \neq 20 - (10 - 5)$

Division

- Changing the order of the terms that you are dividing affects the final answer.

The order of the numbers cannot be rearranged

e.g. $10 \div 5 = 2$
$5 \div 10 = \frac{1}{2}$
$\therefore 10 \div 5 \neq 5 \div 10$

- Changing how numbers being divided are grouped affects its result so the order can't be changed.

The order of the numbers cannot be rearranged

e.g. $(10 \div 5) \div 5 = 2 \div 5 = \frac{2}{5}$
$10 \div (5 \div 5) = 10 \div 1 = 10$
$\therefore (10 \div 5) \div 5 \neq 10 \div (5 \div 5)$

- Dividing by zero is **undefined.**

Anything ÷ zero = UNDEFINED

e.g. $12 \div 0 = undefined$

Identity Elements

- **Identity Element:** Any mathematical object that, when applied by an operation, such as addition or multiplication, to another mathematical object, such as a number leaves the other object **unchanged.**

- The **identity element** for **addition** is zero. When **zero** is added to any number, the number is **unchanged.**

  e.g. $15 + 0 = 15$
- The **identity element** for **multiplication** is one. When **one** is multiplied by any number, the number is **unchanged.**

  e.g. $1 \times 15 = 15$

Basic Operations

A method of addition:

- To add two numbers, you can write one underneath the other. For example, to calculate 452 761 + 38 429, arrange the numbers so that the units line up with units, **tens with tens**, **hundreds with hundreds**, and so on. This helps ensure accurate column addition.

  e.g.
  $$\begin{array}{r} 452\,761 \\ \underline{+\,38\,429} \\ 491\,190 \end{array}$$

Methods of subtraction:

- When we want to subtract numbers from each other we use the **column method.** We need to arrange the numbers so that the **units line up with units**, **tens with tens**, **hundreds with hundreds**, and so on
- When the top number is smaller than the bottom number for a specific column, we can **"borrow"** from the higher column.

  e.g.
  $$\begin{array}{r} 452\,76\mathbf{1} \\ \underline{-\;38\,42\mathbf{9}} \end{array}$$

  Step 1: We need to minus 1 from 9, but 1 is smaller, so we borrow from the 6, which becomes 5, so that 1 can become 11.
  $$\begin{array}{r} 452\,75^{1}\mathbf{1} \\ \underline{-\;38\,42\mathbf{9}} \\ 2 \end{array}$$

  Step 2: Follow this borrow method, if needed, for each column.
  $$\begin{array}{r} 44^{1}\mathbf{2}\;75^{1}1 \\ \underline{-\;3\mathbf{8}\,429} \\ 414\,332 \end{array}$$

A method of multiplication

- To multiply we do it **column for column**, starting at the unit column:

  e.g.
  $$\begin{array}{r} 389 \\ \underline{\times\,15} \end{array}$$

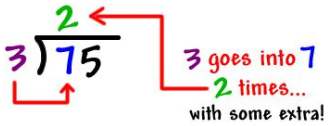
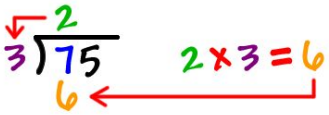
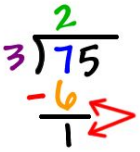
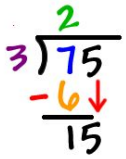
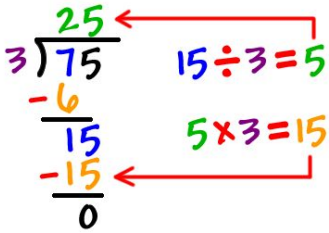
  Step 1: We start with 5
  $$\begin{array}{r} 389 \\ \underline{\times\,15} \\ 1\,945 \end{array}$$
  → product of 389 and 5

Step 2: Now we multiply the tens column, always add a zero far right

$$\begin{array}{r} 389 \\ \times\ 15 \\ \hline 1\,945 \\ +\ 3\,890 \\ \hline 5\,835 \end{array}$$

A process called long division

- **Long division** is a method we use to divide large numbers step by step. It helps us find out **how many times** one number (called the **divisor**) fits into another number (called the **dividend**).
- **Steps:**

Divide:	
Multiply:	
Subtract:	
Bring Down:	
Repeat:	

Inverse Operations

- **Inverse:** reverse in direction or position
- In mathematics, an **inverse operation** is an operation that undo what was done by the previous operation.
  - The inverse of addition is **subtraction**
    e.g. $3 + 2 = 5$
    $5 - 2 = 3$
  - The inverse of subtraction is **addition**

e.g. $3 - 2 = 1$
$1 + 2 = 3$

- The inverse of multiplication is **division**
  e.g. $4 \times 2 = 8$
  $8 \div 2 = 4$
- The inverse of division is **multiplication**
  e.g. $10 \div 5 = 2$
  $2 \times 5 = 10$

- **Additive inverses** are two corresponding positive and negative numbers.
  e.g. 4 and (-4) are additive inverses

Estimations

- We can use two methods to **estimate** answers:
  1. **Doubling and Halving**
     e.g. Estimate the answer to the sum of 5 013 and 4 959.
     Both numbers are closed to 5 000
     $5\,000 + 5\,000 = 10\,000$
  2. **Rounding Off**
     e.g. Estimate the answer to the sum of 3 412 and 6 139.
     Both numbers can be rounded off to the nearest hundred.
     $3\,400 + 6\,100 = 9\,500$

- An **estimate** can also be called an **approximation.**
- The **error** is the difference between the estimate and the actual answer.
  e.g. if we look at the above examples:
  $5\,013 + 4959 = 9\,972$ → actual answer
  $5\,000 + 5\,000 = 10\,000$ → estimated answer
  $\boldsymbol{Error} = 10\,000 - 9\,972 = 28$

Calculators

- When performing certain operations, calculations with large numbers can become difficult and complex.
- We need **calculators** to help us determine the answer to certain calculations.
  e.g. Use a calculator to determine $4\,365 + 7\,388$

#3

Multiples and Factors

Multiples

- A **Multiple** is a number made by multiplying two numbers.

e.g. $10 \times 8 = \mathbf{80}$ — **80** is a multiple of 10 and 8

- We can **list** the multiples of a number by multiplying the given number by an integer.

$$\boxed{\textbf{Multiple of a number} = \textbf{Number} \times \textbf{By any integer (not a fraction)}}$$

- A number may have an **infinite** number of multiples.
  e.g.

Number	First 10 multiples (non-zero)
5	5, 10, 15, 20, 25, 30, 35, 40, 45, 50
7	7, 14, 21, 28, 35, 42, 49, 56, 63, 70
9	9, 18, 27, 36, 45, 54, 63, 72, 81, 90
20	20, 40, 60, 80, 100, 120, 140, 160, 180, 200

LCM

- **Lowest Common Multiple (LCM)** is the lowest multiple shared by two or more numbers.
- **Pattern A:** 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110
- **Pattern B:** 15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165
  1. The numbers in Pattern A are called **multiples of 10**, and those in Pattern B are **multiples of 15**.
  2. Some numbers, such as **30, 60, and 90**, appear in both patterns — these are the **common multiples** of 10 and 15.
  3. The **smallest** of these, which is 30, is called the **Lowest Common Multiple (LCM)** of 10 and 15.

Factors and HCF

- When a number divides evenly into two or more other numbers, it is called a **common factor** of those numbers.
  e.g. **4** is a common factor of **20** and **36** because it divides both numbers exactly.
- The number of **factors** of a number is **finite.**
- A **factor of a number** is always less than or equal to a given number.
  e.g. List all the factors of 30.

$$\frac{30}{1} = 30$$
$$\frac{30}{2} = 15$$
$$\frac{30}{3} = 10$$
$$\frac{30}{5} = 6$$
$$\frac{30}{6} = 5$$

$\frac{30}{10} = 3$
$\frac{30}{15} = 2$
$\frac{30}{30} = 1$
∴ Factors are $30; 15; 10; 6; 5; 3; 2; 1$

- The largest number that divides exactly into two or more other numbers is called their **Highest Common Factor (HCF).**
    1. It is the biggest number that is a factor of all the given numbers.
       e.g. Find the highest common factor of 144 and 240
       Step 1: Write down the factors for each number:
       144: 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144
       240: 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30, 40, 48, 60, 80, 120, 240
       Step 2: Find all the common factors: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48
       Step 3: Find the HCF = 48

Prime Factors

- **Prime factors** are factors of a number that are, themselves, **prime numbers**.
  e.g. The prime factors of 144 are:
  Step 1: Find all the factors first = 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144
  Step 2: Find the prime factors = 2 and 3.

Prime Factorisation

- All the factors in a number can be reduced to **prime numbers.**
    - This is called **reducing** a number to its **prime factors.**
- We can use the following two methods:
    1. **The division ladder method**
    2. **The factor tree method**
- Start with the first prime number, 2, and work through the prime numbers in ascending order, until you have only one remaining and the last prime factor.

Division Ladder Method

- **Example:** Reduce 24 to its prime factors.

24	2
12	2
6	2
3	3
1	

Start with the prime number 2

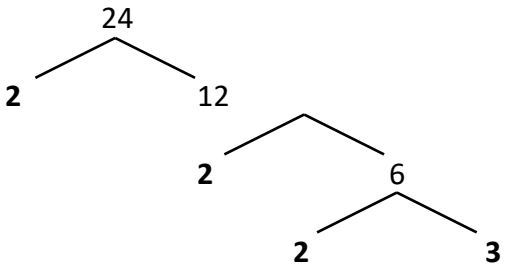
1 is not a prime number

Thus, $2 \times 2 \times 2 \times 3 = 24$

$2^3 \times 3 = 24$

Factor Tree Method

Example: Reduce 24 to its prime factors.



Thus, $2 \times 2 \times 2 \times 3 = 24$

$2^3 \times 3 = 24$

#4

Ratio and Rate

Ratios

- Comparing quantities of the same kind is called a **ratio.**
- It shows the **relationship** between two quantities of the same kind.

  e.g. If James has 5 books and William has 3 books, the ratio of the books they have is **5 : 3**
- Ratios can also be written as a **fraction**

  e.g. the ratio of 5:3 can also be written as $\frac{5}{3}$.
- We can also calculate use the ratio to calculate a specific value.

  e.g. The ratio of girls to boys in the Mathematics class is 8 : 7.If there are 45 students in the class, how many are girls and how many are boys?

  Step 1: First calculate the total number of parts in the ratio

  $8 + 7 = 15$

  Thus, for every 15 learners, 8 are girls and 7 are boys.

  Step 2: Use the ratio of girls and write it as a fraction of 15 to calculate the number of girls in the class.

  $\frac{8}{15} \times 45 = \frac{8}{15} \times \frac{45}{1} = \frac{8}{1} \times \frac{3}{1} = 24$ girls

  Step 3: Repeat step 2 to calculate the number of boys in the class.

  $\frac{7}{15} \times 45 = \frac{7}{15} \times \frac{45}{1} = \frac{7}{1} \times \frac{3}{1} = 21$ boys

Rates

- A **rate** is a special ratio in which the two terms are in different units.
  e.g. If a marathon runner runs a at a speed of 5 minutes per kilometre, time and distance are being compared.
- When you compare **prices,** change ratios to **unit price.**
  e.g. If a 2 kg potatoes cost R50,00, what will a 5 kg potatoes cost?
  $2\text{ kg} = \frac{50}{2} = \text{R}25{,}00$
  $5\text{ kg} = 25 \times 5 = \text{R}125{,}00$

#5

Financial Problem Solving

Value Added Tax

- **Value Added Tax (VAT)** is the form of tax paid by everyone in South Africa.
- **VAT** is now levied at the standard rate of **15%** on the supply of goods and services by registered vendors.
- Basic food, such as bread, milk, fresh fruit and vegetables, etc., are **excluded** from the VAT system.
  - These specific items are **zero-rated and** are usually indicated by the symbol **#.**
- Prices of goods and services are either:
  - **VAT inclusive**
  - **VAT exclusive**
    e.g. The price of an item costs R152,52 excluding VAT. Calculate the VAT-inclusive price.
    Step 1: Calculate the VAT
    $$152{,}52 \times \frac{15}{100} = 22{,}878 = \text{R}22{,}88$$
    Step 2: Calculate the VAT-inclusive amount
    $$152{,}52 + 22{,}88 = \text{R}175{,}40$$
- We can use the following method to calculate the missing value:
  - VAT inclusive percentage = 115%
  - VAT exclusive percentage = 100%
  - VAT percentage = 15%

$$\boxed{\textbf{Missing value} = \textbf{Given Value} \times \frac{\textbf{Unknown Percentage}}{\textbf{Given Percentage}}}$$

e.g. The price of an item costs R152,52 excluding VAT. Calculate the VAT-inclusive price.

$$152{,}52 \times \frac{115}{100} = 175{,}398 = \text{R}175{,}40$$

e.g. The price of an item costs R300,50 VAT inclusive. Calculate the VAT on the price.

$$300{,}50 \times \frac{15}{115} = \text{R}39{,}20$$

Profit and Loss

- **Profit:** The difference between the selling price and the cost price where the **selling price** is **greater** than the cost price.
- **Loss:** The difference between the selling price and the cost price where the **cost price** is **greater** than the selling price.
- Determining the profit or loss:

Given the Amounts	Given the Percentage
• Work out the physical profit or loss • Calculate this as a percentage of the original selling price	• Work out the percentage of the relevant amount, depending on the question • Add or subtract as necessary from the relevant amount

- You compare the cost price and the selling price to determine whether you made a **profit** or **loss.**

Percentage Decrease

- When two numbers are given (original and new), we can calculate the percentage **decrease** between the two numbers:
  1. Work out the difference between the two numbers
  2. Divide the difference (decrease) by the original number and multiply the answer with 100.

     e.g. Jana paid R35,00 for a bag of apples that usually costs R40,00. Calculate the percentage discount she received.

     Step 1: $40 - 35 = 5$

     Step 2: $\frac{5}{40} \times \frac{100}{1} = \frac{500}{40} = 12{,}5\%$

- When the **percentage decrease** and the original number are given for two numbers being compared, we can calculate the new number:
  1. Work out the **percentage** of the amount
  2. Subtract this from the original amount

     e.g. Brian bought shoes that costs R750. When he got to the till, he received 30% discount. How much did he pay for his shoes?

     Step 1: $\frac{30}{100} \times \frac{750}{1} = \frac{22500}{100} = \text{R}225$

Step 2: $750 - 225 = \text{R}525$

Percentage Increase

- When two numbers are given (original and new), we can calculate the percentage **increase** between the two numbers:
    1. Work out the difference between the two numbers
    2. Divide the difference (increase) by the original number and multiply the answer with 100.
       e.g. James earns R6 000 per month. If his employer increased his salary to R6 300 per month, what percentage increase did he receive?
       Step 1: $6\,300 - 6\,000 = 300$
       Step 2: $\frac{300}{6000} \times \frac{100}{1} = \frac{30000}{6000} = 5\%$

- When the **percentage increase** and the original number are given for two numbers being compared, we can calculate the new number:
    1. Work out the **percentage** of the amount
    2. Add this to the original amount
       e.g. Maria invests R1 000. Her investment increased by 3,5%. How much is her investment worth now?
       Step 1: $\frac{3{,}5}{100} \times \frac{1000}{1} = \frac{3500}{100} = \text{R}35$
       Step 2: $1000 + 35 = \text{R}1\,035$

Budgets

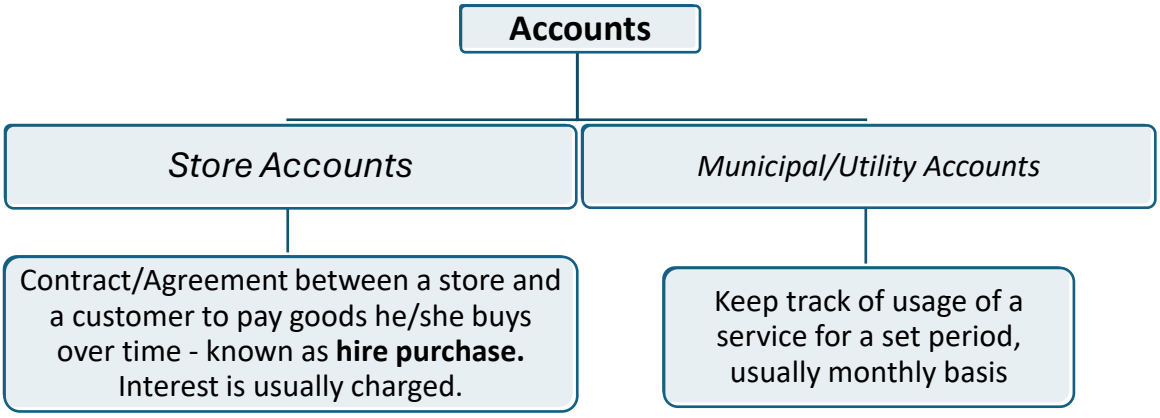
- It is very important to learn how to keep a budget and manage your money.
- Follow these **steps** to draw up a **budget:**
    1. Keep track of and record all your expenses for a month.
    2. Calculate the total family income.
    3. Determine the amount of money available for luxuries or savings by subtracting your expenses from your income.
       e.g. Mr Ndlovu earns R15 000 per month and his wife R12 000 pe month.

Mortgage bond	R7 500
Retirement Annuity	R2 000
Car Repayment Bond	R2 500
Medical Aid	R2 000
Petrol	R1 500
Groceries	R3 500
School Fees	R850
Water and Electricity	R550
Insurance	R1 200
School Bus Tickets	R750

Step 1: Total expenses → $7500 + 2000 + 2500 + 2000 + 1500 + 3500 + 850 + 550 + 1200 + 750 = \text{R}22\,350$
Step 2: $15000 + 12000 = \text{R}27\,000$
Step 3: $27000 - 22350 = \text{R}4\,650$ available for luxuries/savings.

Accounts

- Companies use **accounts** to allow customers to purchase things on credit.
- Accounts are used to keep track of how much money customers owe the company.



Loans

- A **loan** is when you receive money from an authorised financial institution in exchange for future repayment of the principal, **plus interest**.
- The principal is the amount you borrowed, and the interest is the amount charged for receiving the loan.
- The interest is paid according to an **interest rate**.
- An **interest rate** is quoted as a **percentage** of the loan, for example, 15%.

  e.g. Ms Davids borrowed R 5 000 from W-Bank at an interest rate of 10% per year. What is the total amount that Ms Davids will have to pay back after one year?

  Step 1: Calculate percentage interest → $5000 \times \frac{15}{100} = \text{R}750$

  Step 2: Add the interest amount to the loan amount → $5000 + 750 = R5\ 750$

Simple Interest

- Interest needs to be paid when a loan is taken from a bank.
- Interest can also be earned when investing money.
- **Simple interest** refers to the money charged or earned at a constant rate.

  e.g. Calculate the amount of interest on R850 at a rate of 15% per annum.

  $850 \times 15\% \times 1$

  $= \frac{850}{1} \times \frac{15}{100} \times \frac{1}{1}$

  $= \frac{12\ 750}{100}$

  $= \text{R}127{,}50$

e.g. Mr. Dlamini borrows R60 000 from XYZ Bank at an interest rate of 8,75% per annum. He takes the loan for three years. What is the total amount of Mr. Dlamini's monthly instalment on the loan?

Step 1: Let's calculate the total amount of the loan, interest included.

$60\ 000 \times 8{,}5\% \times 3$

$= \frac{60\ 000}{1} \times \frac{8{,}5}{100} \times \frac{3}{1}$

$= \frac{1\ 550\ 000}{100}$

$= \text{R}15\ 300$ interest

$60\ 000 + 15\ 300 = \text{R}75\ 300$ total amount to repay over 3 years.

Step 2: Now calculate the monthly instalment

$75\ 300 \div (3 \times 12)$

$= \frac{75\ 300}{36}$

$\text{R}2\ 091{,}67$ per month for three years

Hire Purchase

- **Hire purchase agreements** are used as an arrangement when purchasing goods or services.
- The purchaser will pay the initial instalment or down payment at the beginning, followed by additional payments in the future to pay off the remaining balance of the goods, plus interest.

  e.g. You buy a television (TV) at Electronics Supply for R6 999. You can agree to pay the TV off over 14 months to make it more affordable. The amount you will pay, free of interest, is:

  $\frac{6\ 999}{14} = \text{R}499{,}93$ without interest.
- Hire purchase agreements usually require **interest** to be paid.

Exchange Rates

- An **exchange rate** (or foreign exchange rate) between two currencies is the rate at which you can exchange one currency for another.
- It's the **exchange rate** at which one country's currency can be exchanged for another country's currency.

  e.g. An exchange rate of approximately R24,38 South African Rand (ZAR) to the British Pound Sterling (GBP) means that you will pay R24,38 for £1 or that for £1 you will get R24,38.
- **Method:**
  1. Identify the currency you have (what you are starting with).
  2. Identify the currency you want to determine (what you are converting to).
  3. Multiply the amount you want to determine by a fraction:

$$\boldsymbol{New\ Currency = Amount \times \frac{Currency\ you\ WANT}{Currency\ you\ HAVE}}$$

e.g. Kabelo and his wife are planning a holiday to Australia. They estimate that they will need approximately $8 500 AUD to cover all expenses. If the exchange rate is R12,42 for $1 AUD, how many rands must they exchange to get the full amount?

$$8\,500 \times \frac{12{,}42}{1} = \text{R}105\,570$$

Time to make some notes:

2. Integers

#1

Counting, Ordering and Comparing Integers

Integers

- **Integers (ℤ)** include whole numbers which are both positive or negative or zero.
  ℤ = {... -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6 ...}

  On a number line, integers look like this:

- An **integer** is a number with no fractional part or decimal, from the set of negative and positive numbers, including zero.
  e.g. -7; -3; 0; 2 ; 9; 305 ; …
- **Integers** extend from **negative infinity** to **positive infinity.**
  - This means that the numbers on the number line continue **indefinitely** in both directions
- **Infinity** is indicated on a number line by arrows indicating the line continues in both directions.

Negative Numbers

- A **negative number** is a number that is **less than zero.**

- The **negative sign** tells you how far away the number is from zero.
  - So, -2 means you are two steps away from 0 and -4 means you are 4 steps away from 0.
  - Thus, -4 is smaller than -2, because it is further way from zero.

Counting

- When you **count** in **integers,** you are counting the whole numbers above and below zero.
- **Number lines** can also be used for counting.

- You can also count in integers without a number line, using a **number pattern.**

  2; 4; 6; 8; 10; 12; …

Comparing

- We use the following **symbols** when comparing integers.

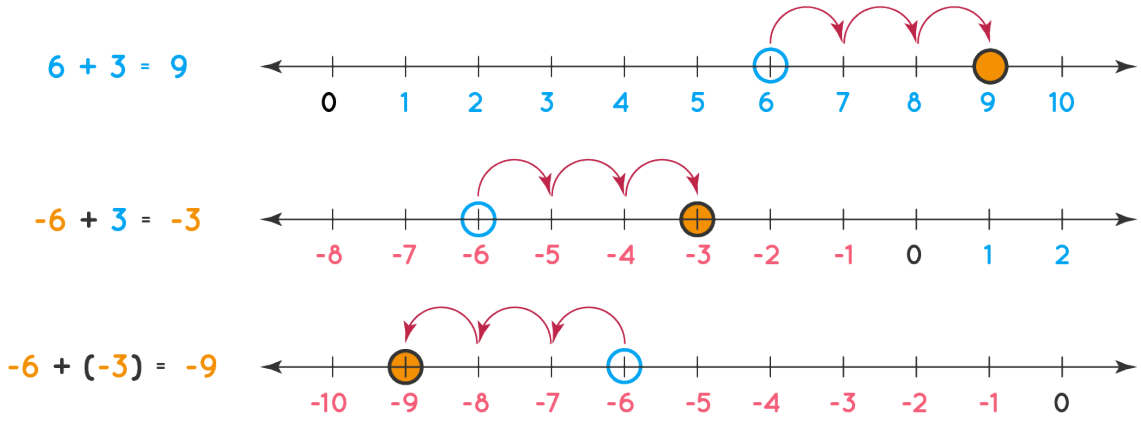
Symbol	Meaning	Example
=	Equal to	$15 + 13 = 28$
>	Greater than	$-23 > -150$
<	Smaller than	$-7 < 3$

#2

Adding and Subtracting Integers

Adding Integers

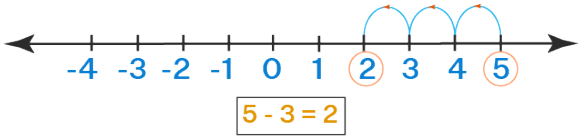
- A number line is very useful when **adding integers.**
- When you add a positive (+) integer: *Move right on the number line.*
- When you add a negative (−) integer: *Move left on the number line.*
    - Adding a negative number has the same effect as **subtracting** a natural number.
- When you represent **integers** n expressions, you can put **brackets** around the numbers to help isolate the integer from the operation.
    - When you add **positive integers,** the signs are simple.



Subtracting Integers

- A number line is very useful when **subtracting integers.**
- When you subtract a positive (+) integer: *Move left on the number line.*
- When you subtract a negative (−) integer: *Move right on the number line.*

- When you **subtract** negative inverses, the sign becomes more complicated.
  - Subtracting a negative number has the same effect as **adding** a natural number.
- **Take note:**
  - Subtracting a negative (−) integer is the same as adding its **additive inverse**.
  - The additive inverse of −5 is +5 and the additive inverse of −3 is +3.



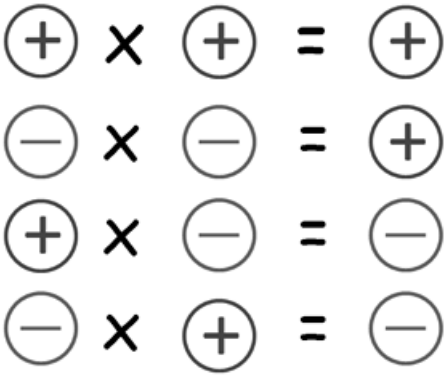
#3

Multiplying and Dividing Integers

Multiplying Integers

- Multiplication of integers is **commutative.**

  e.g. $5 \times (-10) = (-10) \times 5$
- With integers, **multiplication** distributes over addition and subtraction.



When multiplying or dividing integers, remember these sign rules:

- **Same signs** (both positive or both negative) give a **positive** answer.
- **Different signs** (one positive and one negative) give a **negative answer.**

e.g. $5 \times 10 = 50$
$-5 \times (-10) = 50$
$5 \times (-10) = -50$
$-5 \times 10 = -50$

- Multiplication with a **positive number** distributes over addition and subtraction integers.

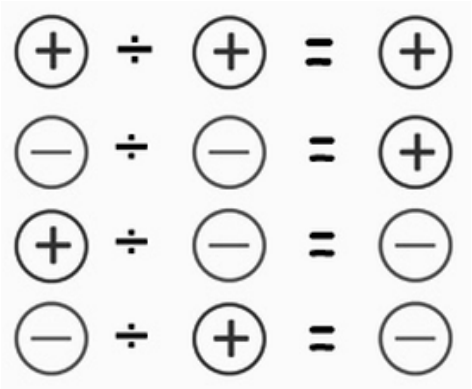
  e.g. $10 \times \big(10 + (-2)\big) = 10 \times (10 - 2) = 10 \times 8 = 80$
  $10 \times 10 + 10 \times (-2) = 100 - 20 = 80$

  e.g. $10 \times \big(10 - (-2)\big) = 10 \times (10 + 2) = 10 \times 12 = 120$
  $10 \times 10 - 10 \times (-2) = 100 + 20 = 120$

Dividing Integers

- Division is the **inverse** of multiplication.
- An expression that specifies division only, such as $80 \div 8$, is called a **quotient** or a **quotient expression.**

  e.g. $80 \div 8 = 10$ → 10 is called the **quotient** of 80 and 8.



e.g. $10 \div 5 = 2$
$-10 \div (-5) = 2$
$10 \times (-2) = -2$
$-10 \div 5 = -2$

- The following rules always apply:

Any non-zero number ÷ zero = UNDEFINED

Zero ÷ non-zero number = 0

- You can write any division as a **fraction.**

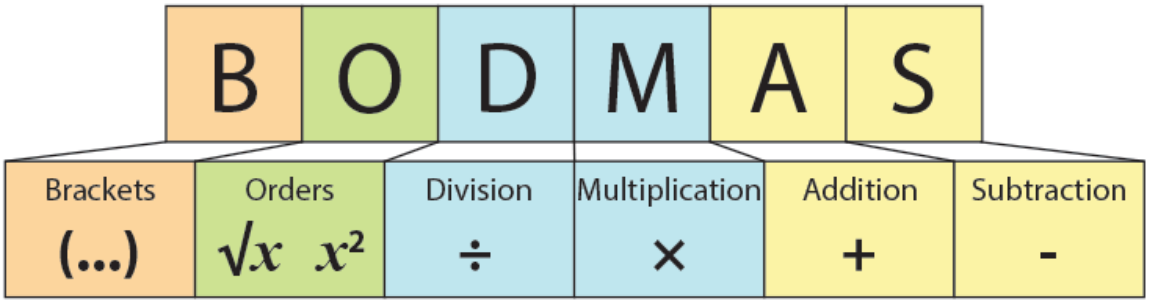
  e.g. $10 \div 2 = \frac{10}{2}$

#4

Multiple Operations with Integers

Mixed Calculations

- Order of operations:



e.g. $-10 \times (-5) + (-20) \div (-2)$
$= [-10 \times (-5)] + [(-20) \div (-2)]$
$= 50 + 10$
$= 60$

Squares and Cubes

- A **square number** is the result when a number is multiplied by itself.
  - It is the product of two identical factors
    e.g. $4 \times 4 = 4^2 = 16$

- To **cube** a number is to multiply a number by itself and then multiply it again by itself.
  - It is the product of three identical factors
    e.g. $2 \times 2 \times 2 = 2^3 = 8$

Square Roots

- To find the **square root** of a number, you want to find the number, that when **multiplied by itself**, gives you the original number.
  - It is another number that, when squared, will equal the first number
  - Square root is notated by $\sqrt{}$
    e.g. 5 is the square root of 25, because if we multiply 5 by itself (5×5), we get 25. Thus, $\sqrt{25} = 5$
- Square roots can be **positive** or **negative.**
  e.g. The square root of 100 is 10 and -10
  10 → **positive square root**
  -10 → **negative square root**

Cube Roots

- The **cube root** of a number is the reverse process of the cube of a number.
- To find the **cube root** of a number, you want to find the number, that when **multiplied by itself and then multiplied by itself again**, gives you the original number.
  - It is another number that, when cubed, will equal the first number
  - Square root is notated by $\sqrt[3]{}$

e.g. $\sqrt[3]{27} = 3$
e.g. $\sqrt[3]{-27} = -3$

#5

Properties of Integers

Addition and subtraction are **inverse operations** of one another.

Commutative Properties

- The **commutative property** states that the numbers on which we operate can be moved or swopped from their position without making a difference to the answer.
  e.g. $30 + 40 = 40 + 30 = 70$
- We can apply the commutative property on **addition** and **multiplication.**
  e.g. $3 \times 4 = 4 \times 3 = 12$
- The commutative property of **addition** is:

$$\boxed{\boldsymbol{a + b = b + a}}$$

- The commutative property on **multiplication** is:

$$\boxed{\boldsymbol{a \times b = b \times a}}$$

Commutative Properties of Integers

- Integers for **addition:**

(+) + (+) = (+)
(-) + (-) = (-)
(+) + (-) = Use the sign from the bigger number
(-) + (+) = Use the sign from the bigger number

  e.g. $-2 + (-3) = (-3) + (-2) = -5$

- Integers for **multiplication:**

$$+ \times + = +$$
$$- \times - = +$$
$$+ \times - = -$$
$$- \times + = -$$

Associative Property

- The **associative property** states that when three or more numbers are added or multiplied, the sum or the product is the same regardless of the grouping of the addends or multiplicands.

  e.g. $9 + (2 + 5) = (9 + 2) + 5 = 16$

  e.g. $9 \times (2 \times 5) = (9 \times 2) \times 5 = 90$
- The associative property can only be used with **addition** and **multiplication.**

The Associative Property of Integers

- The **associative property** is the property of **grouping.**

  e.g. $(-9) + (-2) + (-8)$

  Grouping option 1:

  $= [(-9) + (-2)] + (-8)$

  $= (-11) + (-8)$

  $= -19$

  Grouping option 2:

  $= (-9) + [(-2) + (-8)]$

  $= (-9) + (-10)$

  $= -19$

The Distributive Property of Integers

- The **distributive property** of integers can be stated as the product of an integer with the sum of two integers inside the parentheses is equal to the sum of the products of integers separately.

e.g.	$-5[4 + (-3)]$	or	$-5[4 + (-3)]$
	$= [(-5) \times 4] + [(-5 \times (-3)]$		$= -5(1)$
	$= (-20) + 15$		$= -5$
	$= -5$		

Time to make some notes:

3. Exponents

#1

Squares and Cubes

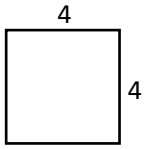
Square Numbers

- A **square number** is the result when a number is multiplied by itself.
- If you multiply any **identical numbers,** you will find the area of a square with side length that number.

  e.g. $4 \times 4 = 4^2 = 16$

  4×4 → **Expanded notation**

  4^2 →**Exponential notation**



Square Roots

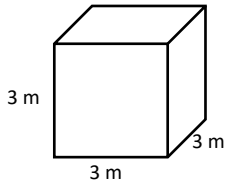
- To find the **square root** of a number, you want to find the number, that when **multiplied by itself**, gives you the original number.
  - Square root is notated by $\sqrt{}$

    e.g. 5 is the square root of 25, because if we multiply 5 by itself (5×5), we get 25. Thus, $\sqrt{25} = 5$

Cube Numbers

- To **cube** a number is to multiply a number by itself and then multiply it again by itself.
- If you multiply any number by itself 3 times, you will get the inside volume of a cube, whose side lengths are equal to that number.

  e.g. $3 \times 3 \times 3 = 3^3 = 27$



Cube Roots

- The **cube root** of a number is the reverse process of the cube of a number.
- To find the **cube root** of a number, you want to find the number, that when **multiplied by itself and then multiplied by itself again**, gives you the original number.
  - Square root is notated by $\sqrt[3]{}$

    e.g. $\sqrt[3]{27} = 3^3 = 3 \times 3 \times 3$

Integers

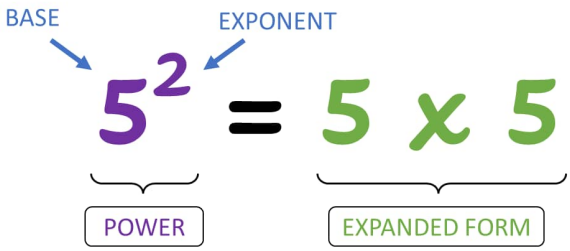
- The calculator "understands" -6^2 and $(-6)^2$ as two different numbers.
  - It understands -6^2 as $-6 \times 6 = -36$
  - It understands $(-6)^2$ as $-6 \times -6 = 36$

#2

Laws of Exponents

The Exponential Form

- **Exponential Notation** is used to represent a square number.



- To represent a number in **exponential form**, you can just calculate how many times it is **multiplied** by itself.
    - The **base** is the original number
    - The **exponent** is the number of times it has been multiplied by itself
      e.g. $3 \times 3 \times 3 \times 3 = 3^4$
- The **index** is an exponent that indicates the number of factors that are multiplied.
- When two **different numbers** are multiplied by itself in **expanded notation,** you simply keep them separate and count how many times each number is multiplied by itself.
  e.g. $3 \times 3 \times 2 \times 2 \times 2 = 3^2 \times 2^3$

The Power of 1

- When 1 is the exponent, it has a unique characteristic.
  e.g. 4^1 means that the number 4 is not multiplied by itself at all.
- Anything to the power of 1 is equal to the number **itself.**

$$\boxed{x^1 = x}$$

- The answer of 1 multiplied by itself any number of times will always be equal to **1**.

$$\boxed{1^n = 1}$$

Product of Powers

- The **product of powers** with the same base is equal to the same base, raised to the sum of the exponents of the powers.

$$\boxed{a^m \times a^n = a^{m+n}}$$

e.g. $4^2 \times 4^3 = 4^{2+3} = 4^5$

Power of a Power

- A **power raised to a power** is equal to the same base, raised to the product of the exponents.

$$\boxed{(a^m)^n = a^{m \times n}}$$

e.g. $(8^2)^3 = 8^{2\times3} = 8^6$

Power of a Product

- The **power of a product** is equal to the product of the factors, each raised to the same power.

$$\boxed{(a \times b)^m = a^m \times b^n}$$

e.g. $(8 \times 2)^6 = 8^6 \times 2^6$

Quotient of Powers

- The **quotient of powers** with the same base is equal to the same base, raised to the difference between the exponents of the powers.

$$\boxed{a^m \div a^n = a^{m-n}}$$

e.g. $8^6 \div 8^4 = 8^{6-4} = 8^2$

#3

Mixed Operations and Scientific Notation

Negative Numbers

- A **negative number** raised to a power must be inserted in brackets.
  e.g. $(-5)^2 = (-5) \times (-5) = 25$
- When a **negative integer** is raised to an even power, the answer is positive.
  e.g. $(-7)^2 = (-7) \times (-7) = 49$
- When a **negative integer** is raised to an odd power, the answer is negative.
  e.g. $(-4)^3 = (-4) \times (-4) \times (-4) = -64$

Square Roots

- Since a square root sign is the same as a bracket, according to the BODMAS rule, the operation in the square root must be done first.

e.g. $\sqrt{44+20} + \sqrt{9}$
$= \sqrt{64} + \sqrt{9}$
$= 8 + 3$
$= 11$

Cube Roots

- Since a cube root sign is the same as a bracket, according to the BODMAS rule, the operation in the cube root must be done first.

e.g. $\sqrt[3]{38-11} + \sqrt[3]{8}$
$= \sqrt[3]{27} + \sqrt[3]{8}$
$= 3 + 2$
$= 5$

Fractions

- **Squaring or cubing** a fraction or a decimal fraction is no different from squaring or cubing an integer.

  e.g. $\left(\frac{5}{7}\right)^2 = \frac{5}{7} \times \frac{5}{7} = \frac{5^2}{7^2} = \frac{25}{49}$

  e.g. $\left(\frac{3}{5}\right)^3 = \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} = \frac{3^3}{5^3} = \frac{27}{125}$
- The **square root** of a common fraction can be calculated by the square root of the numerator divided by the square root of the denominator.

  e.g. $\sqrt{\frac{9}{16}}$ → $\frac{9}{16}$ is $\frac{3}{4} \times \frac{3}{4}$

  $= \frac{\sqrt{9}}{\sqrt{16}}$

  $= \frac{3}{4}$

Scientific Notation

- **Scientific notation** is a way of writing very large or very small numbers in a shorter form using powers of 10.
- The format is:

$$\boxed{a \times 10^n}$$

  where:
  - a is a number between 1 and 10
  - n is an integer (can be positive or negative)

  e.g. large number: 300 000 = 3×10^5

  small number: 0.00042 = 4.2×10^{-4}

Time to make some notes:

4. Numeric and Geometric Patterns, Functions and Relationships

#1

Investigate and Extend Patterns

Number Patterns

- A **number pattern** is a sequence of numbers that follow a specific rule.
- In some cases, each number in a **sequence** can be formed from the previous number by performing a specific function or operation.
  e.g. 3; 8; 13; 18; 23; … → 5 is added every time
- A **sequence** can be formed by repeatedly multiplying or dividing by the same number.
  e.g. 5; 10; 20; 40; 80; … → the ratio between the one term and the next is constant. Every term is multiplied by 2 to get the next term
- A **sequence** can also be formed in such a way that neither the difference nor the ratio between one term and the next is constant.
  e.g. 3; 5; 9; 15; 23; … → consecutive even numbers, starting with 2, are added to form the next term

Using Tables

- A **number pattern** can be represented in table form.
  e.g. 1; 5; 9; 13; 17; …

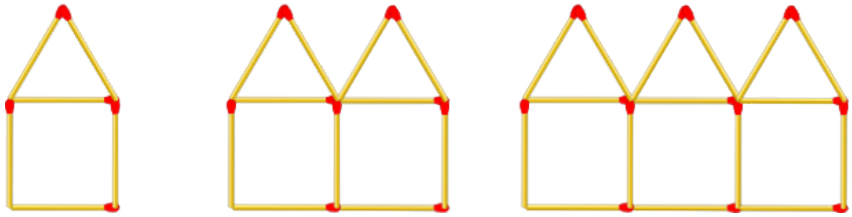
Position	1	2	3	4	5
Term	1	5	9	13	17

  - In the top row of the table, we write the position in the pattern
  - In the bottom/second row of the table, we write the term in the pattern

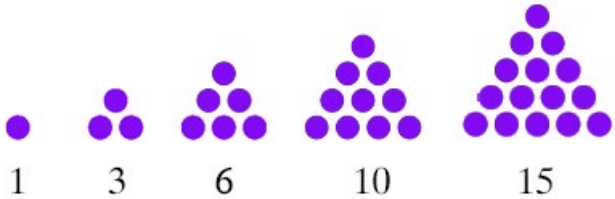
- The **table** helps in keeping track of all the terms' positions in the pattern.
- Tables can also help you see and **extend** the pattern.

Geometric Patterns

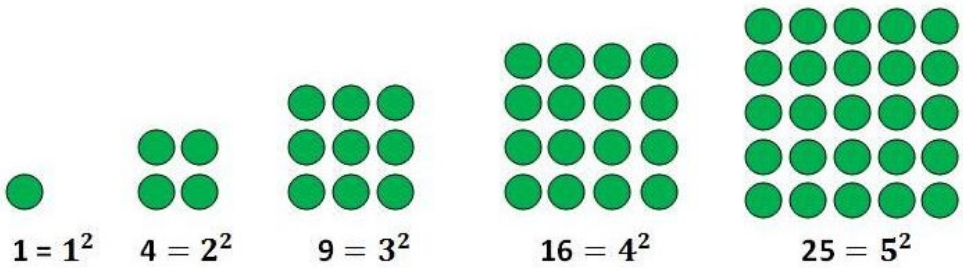
- **Geometric Patterns** consist of a series of shapes.
- Patterns made from shapes are similar to patterns made from numbers because the pattern is determined by a rule.
- **Geometric Patterns** can range from simple shape patterns, such as tessellations, to more complex patterns, such as dot patterns.
    - A **tessellation** is when we cover a surface with a pattern of flat shapes so that there are no overlaps or gaps.
      e.g.



- A **triangular number** is a number that can be represented by a pattern of dots arranged in an equilateral triangle with the same number of dots on each side



- We can represent numbers as an arrangement of dots or circles in various shapes, for example triangles, squares, pentagons, hexagons, and so on.
    - Those numbers are called **polygonal numbers**



- A **square** is a number that is obtained by multiplying a number by itself.

#2

The General Rule

Negative Numbers

- Negative numbers can be part of **patterns.**
  e.g. -2; 0; 2; 4; 6; …

Finding the Rule

- It is important that you can find the rule for a pattern and use this rule to find any term in the sequence.
  e.g. 3; 6; 9; 12; 15; …
  Step 1: Arrange the pattern into a table of input and output values

Input value	1	2	3	4	5
Output value	3	6	9	12	15

  Step 2: We use trial and improvement to work out the rule

How can we get 3?	*How can we get 6?*	*How can we get 9?*
$1 \times 1 + 2$	$2 \times 1 + 4$	$3 \times 1 + 6$
$1 \times 2 + 1$	$2 \times 2 + 2$	$3 \times 2 + 3$
$\underline{\mathbf{1 \times 3}}$	$\underline{\mathbf{2 \times 3}}$	$\mathbf{3 \times 3}$
$1 \times 4 - 1$	$2 \times 4 - 2$	$3 \times 4 - 3$

  Thus, Rule: **Term Number** $\times$ $\mathbf{3}$

Using the Rule

- We can use the **rule** to calculate missing values:
  - Term number
  - nth term

  e.g. 3; 6; 9; 12; 15; …

Input value	1	2	3	4	5
Output value	3	6	9	12	15

  Rule: **Term Number** $\times$ $\mathbf{3}$

  1. Use the rule to determine the value of the 100th term
     Term Number $\times 3 = 100 \times 3 = 300$
  2. Use the rule to determine the term number which has a value of 168
     Term Number $\times 3 = 168$
     Term number $= \frac{168}{3} = 56$

Formula

- The formula to calculate the nth term in an arithmetic sequence is:

$$\boxed{Tn = a + (n - 1)d}$$

  - $\boldsymbol{a}$**:** The 1st term
  - $\boldsymbol{d}$**:** Common difference

e.g. Determine the general rule for the sequence below:

1; 4; 7; 10; 13

Step 1: Substitute

$Tn = a + (n - 1)d$

$Tn = 1 + (n - 1)(3)$

Step 2: Multiply

$Tn = 1 + 3n - 3$

$Tn = 3n - 2$

#3

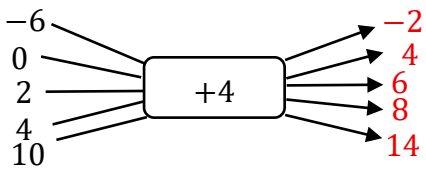
Input and Output Values

Functions

- A **function** relates an **input** to an **output.**
- A **function** is the process that allows you to take a number, as an **input,** and produce a single number as an **output.**
- If one variable quantity is influenced by another, we say there is a **relationship** between the two variables.
- It is sometimes possible to find out what value of the one quantity, in other words what number, is linked to a specific value of the other quantity.
- Relationships and **functions** can be described using words, flow diagrams, tables, or formulae.

Flow Diagrams

- Each **input number** in a flow diagram has a corresponding **output number.**
- The first (top) input number corresponds to the first (top) output number.



Input numbers **Output numbers**

e.g. $5 \longrightarrow \boxed{n^2 - 4} \longrightarrow 21$

$= 5^2 - 4$

$= 25 - 4$

$= 21$

Tables

- The top row contains the **input values,** while the bottom row contains the **output values.**

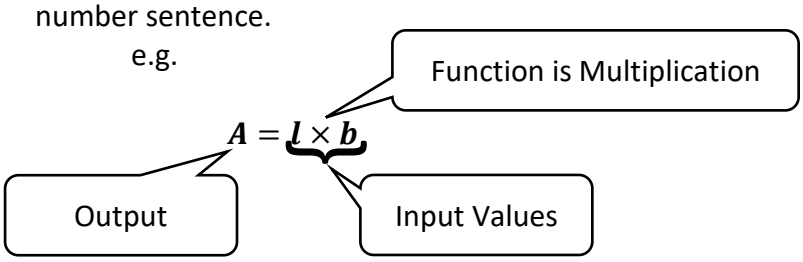
Input	1	2	3	5	10
Output: $\times$ **3** $+$ **7**	10	13	16	22	37

$1 \times 3 + 7 = 10$
$2 \times 3 + 7 = 13$
$3 \times 3 + 7 = 16$
$5 \times 3 + 7 = 22$
$10 \times 3 + 7 = 37$

Using a Formula

- In mathematics, **rules** can be followed to produce certain results. Each input will have a specific output based on the rule.
- **Input values** are the numbers used in the sentence, while the **output** is the result given by the formula.
- The **function or rule** is represented by the mathematical operation used in the number sentence.

  e.g.



- **Input Numbers** can be any type of number while **Output Numbers** depend on the input numbers as well as the operations/function used.

  e.g. Write a formula for the following: The input can be represented by the letter x, the output can be represented by the letter y and the rule/function is $\times$ 4.

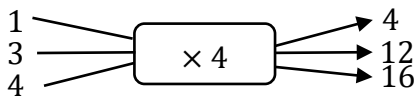
  $y = x \times 4$

Equivalent Forms

Relationships can be represented by any of the following forms:

- **Verbal**
- **Flow diagrams**
- **Tables**
- **Formulae**
- **Number Sentences**

  e.g. Represent $y = x \times 4$ in a flow diagram and a table for the values 1,3 and 4.



Input values	1	3	4
Output values	4	12	16

Time to make some notes:

5. Algebraic Expressions

#1

Conventions of Algebra

Algebra

- An **algebraic expression** is a symbolic description of a set of calculations that can be performed on different values of a variable.

  e.g. $2x + 5$

  This expression means multiply the value of x by 2 and add 5 to the answer

- A **formula** is a description of how the values of a variable, that depends on one or more other variables, can be calculated.

  e.g. $y = 2x + 5$

  This formula means to calculate the value of y that corresponds to a specific value of x, multiply the value of x by 2 and add 5

- An **equation** is a statement about an unknown number. An equation states what result (answer) is obtained if certain calculations are performed on an unknown constant.

  e.g. $2x + 5 = 11$

  This equation states that when a certain unknown number (x) is multiplied by 2 and three is added to the result, the answer is 11

Algebraic Conventions

- Wherever possible, write the **multiplication** of factors without a multiplication sign.
  - Always write the numerical coefficient in front

    e.g. $x + x = 2 \times x = 2x$
- Repeated multiplication of a number by itself can be written in **exponential form**.

  e.g. $x \times x \times x = x^3$
- If the numerical coefficient in front of a variable is **1**, we do not write/show it.

  e.g. $3x - 2x = x$
- Wherever possible, write **division** as a **fraction.**

  e.g. $x \div y = \frac{x}{y}$
- Arrange the variables **alphabetically** when a term has more than one variable.

  e.g. $4abc$ and not $4bac$

- An **equation** means that the number sentence has an **equal sign.**
- If the number sentence **doesn't** have an **equal sign,** it is called an **expression.**
- **Expressions** and **equations** consist of **constants** and **variables.**
  - A **variable** is usually a letter that acts as a **place holder** for a value
  - A **constant** refers to a value that will **not change**, even if the variable changes

    e.g.

$$y = 3x - 6$$

The **variables** are x and y

It has an equal sign so is an **equation**

The **constant** is 6

#2

Like an Unlike Terms

- **Terms:** Different parts of an expression.
  e.g. $3x - 2y + 4z$
  $3x;\ 2y; 4z$ are the terms
- **Terms** are separated by **plus** and **minus** signs.
- **Terms** can contain products.
- An expression can have **like terms, unlike terms,** or **both.**
  - **Like terms** are terms that have the same variable(s) raised to the same power.
    e.g. $3x + 7x$ or $3x^2 + 4x^2$
  - **Unlike terms** are two or more terms that are not like terms, i.e. they do not have the same variables or powers.
    e.g. $3x + 5x^2$

- The number of terms in an expression determines how the expression is named.
  - **Mononomial:** An algebraic expression with exactly one term
  - **Binomial:** An algebraic expression with exactly two terms
  - **Trinomial:** An algebraic expression with exactly three terms
  - **Polynomial:** An algebraic expression with many term

- A **term** is made up of two coefficients:
  - One **numerical**
  - One **alphabetical**

#3

Coefficients and Constants

Coefficients

- A **term** is made up of two coefficients:
  - One **numerical**
  - One **alphabetical**
- The **alphabetical** coefficient can consist of more than one variable, some which can be written in exponential form.
  e.g. $3xy^2$
  3 → numerical coefficient
  xy^2 → alphabetical coefficient, consisting of two variables, x and y^2
- In algebra, the **sign** belongs to the numerical variable.
  e.g. $5a - 2b$ → the **minus** sign belongs to 2
- When like terms are added, we add the **numerical coefficient** of the terms.

Exponents

- An **exponent** in a term only applies to the factor that comes before it.
- When a bracket is raised to a power, all the components inside it, including the numerical coefficients, are also raised to the power.

  e.g. $(3x)^2 + 2y = 3^2x^2 + 2y$

#4

Adding and Subtracting Algebraic Equations

Laws and Expressions

- **Commutative Property:** the order in which we add or multiply numbers does not change the answer.

  e.g. a + b = b + a

  e.g. ab = ba
- **Associative Property:** the way in which we group three or more numbers when adding or multiplying does not change the answer.

  e.g. (a + b) + c = a + (b + c)

  e.g. (ab)c = a(bc)
- **Distributive Property:** Multiplication can be distributed over addition and subtraction.

  e.g. a(b + c) = ab + ac

Equivalent Expressions

- We can find an **equivalent expression** by rearranging and combining like terms.
- Like terms can combine to form a single term.

  e.g. the terms $30x$ and $10x$ can combine to form $40x$

  e.g. $3x + 15 - x + 12 + 4x$ can become $6x + 17$

#5

Equations

- An **equation** is a mathematical sentence that is true for some numbers but false for other numbers.

e.g. $x + 5 = 8$ is true for $x = 3$, but false for $x = 6$

- Two or more equations can have the same solutions.
  - They are then called **equivalent equations.**
    e.g. $x + 3 = 13$ and $2x = 20$

Time to make some notes:

END OF TERM 1 STUDY NOTES